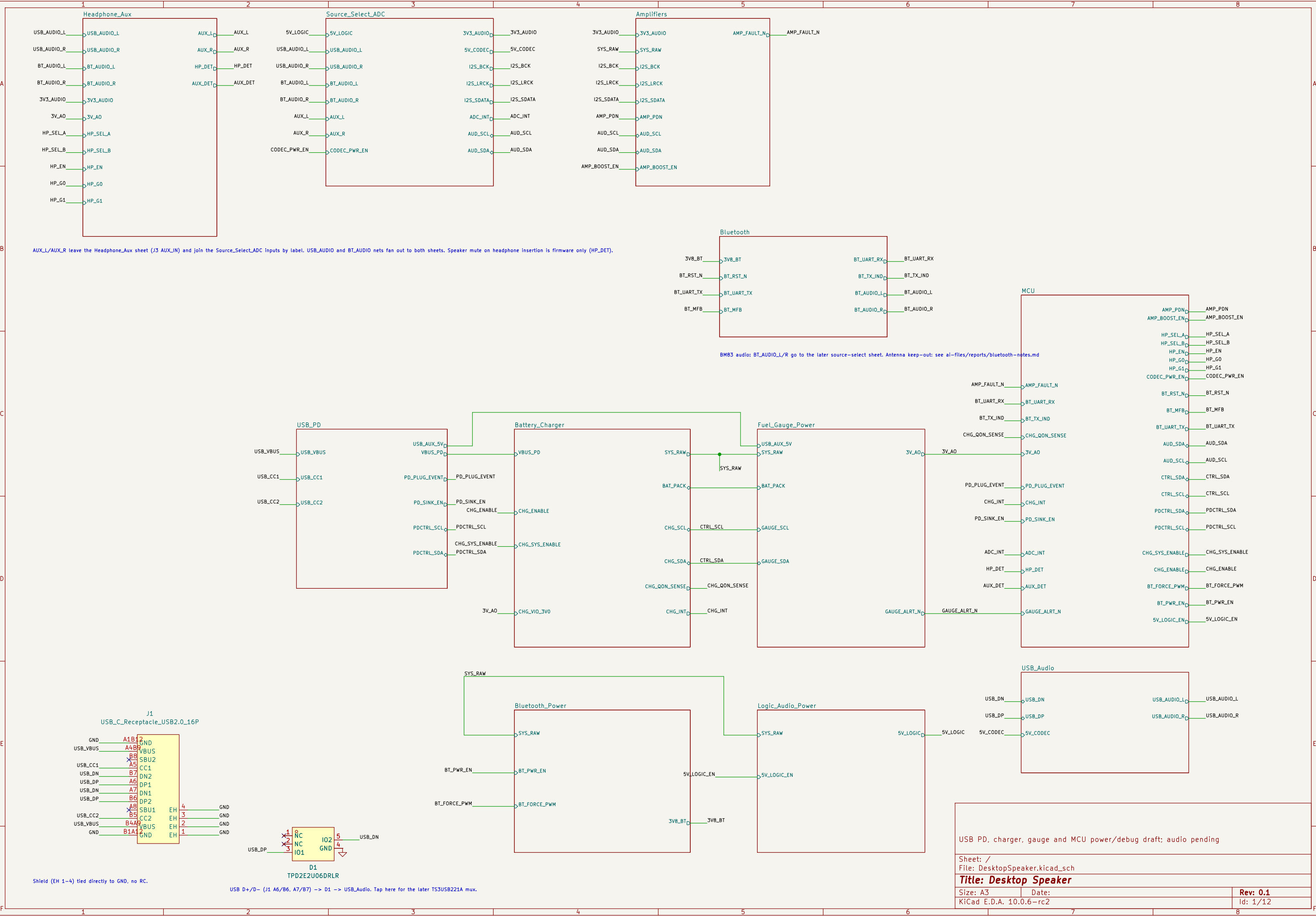
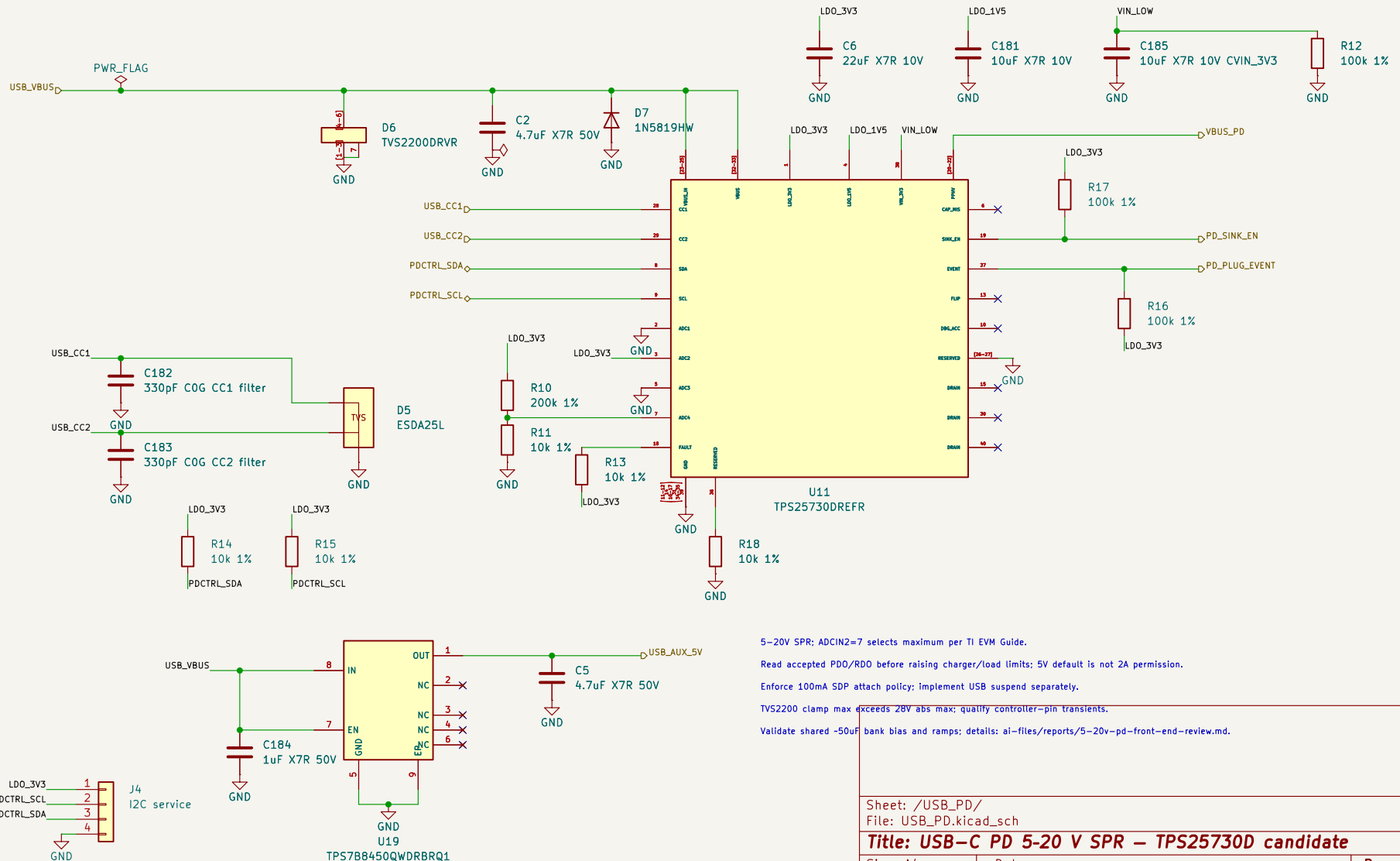
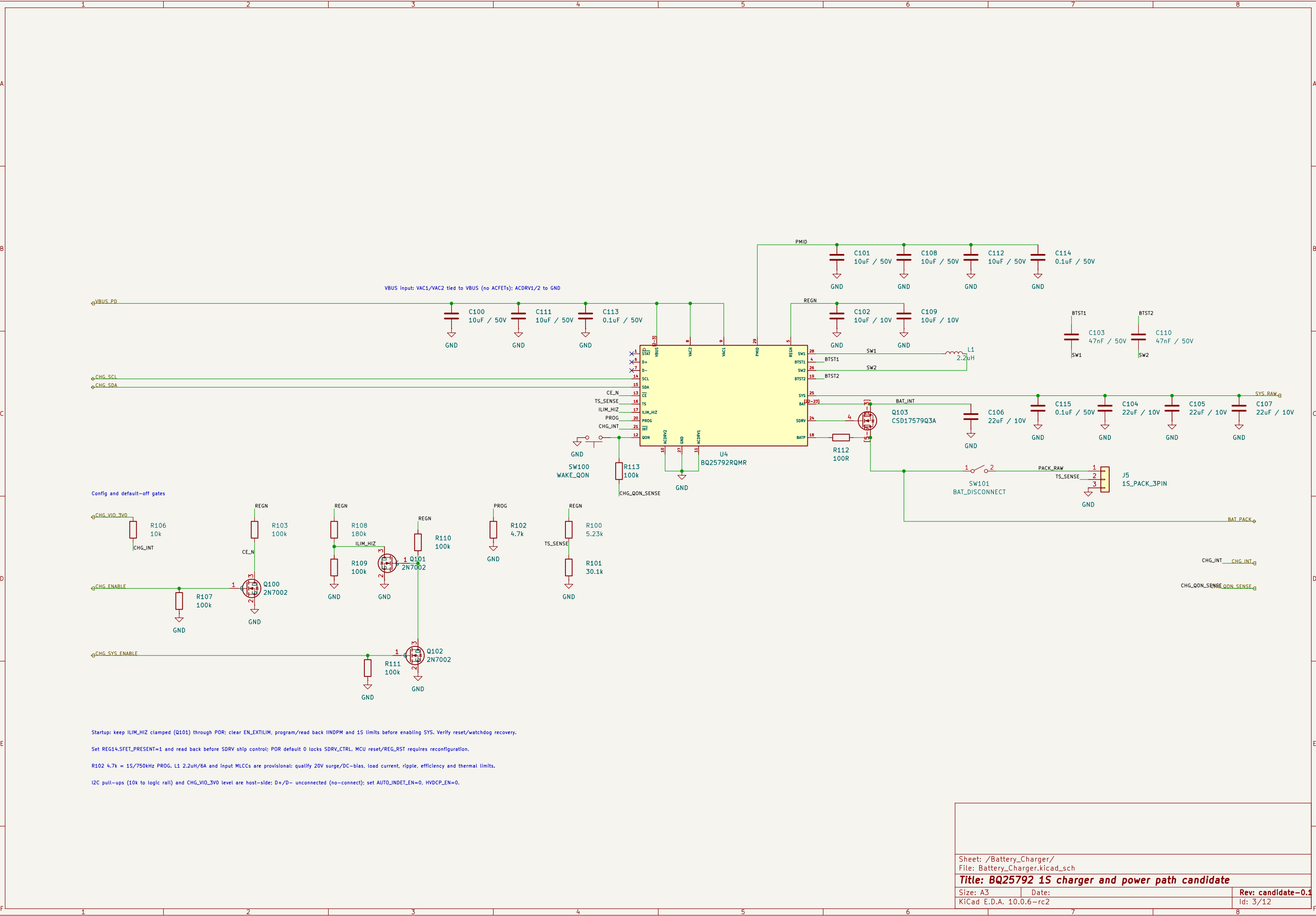
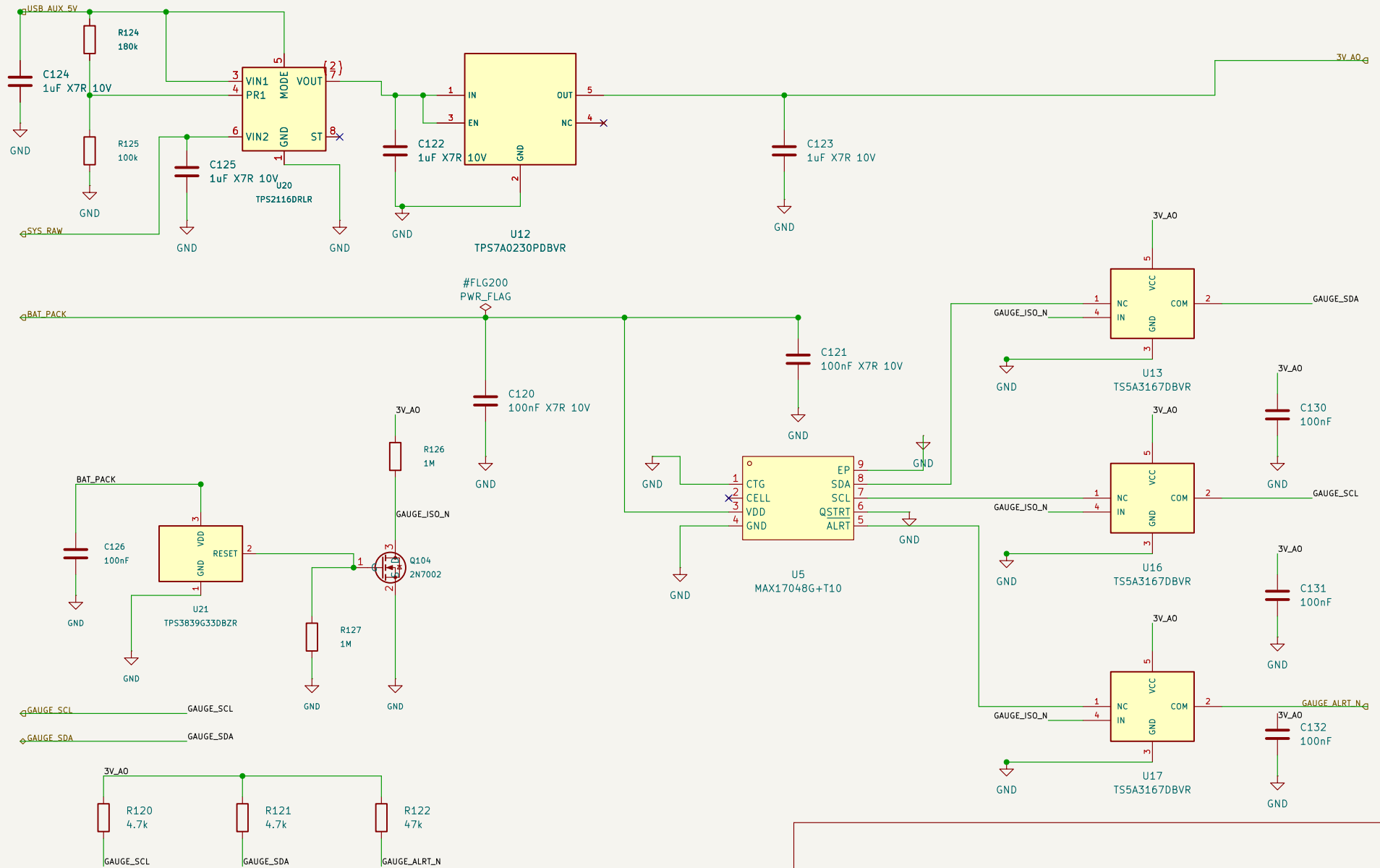










USB—priority logic supply; PR1 nominal 2.8 V.
Switchover/BOR and effective capacitance need qualification.

CELL pad 2 unused on MAX17048; QSTRT tied low

Gauge bus; AO—powered switches; default OFF below valid battery.
U21 3.08 V supervisor enables Q104 after reset recovery.

Sheet: /FuelGauge_Power/
File: FuelGauge_Power.kicad_sch

Title: Fuel gauge and low-current power

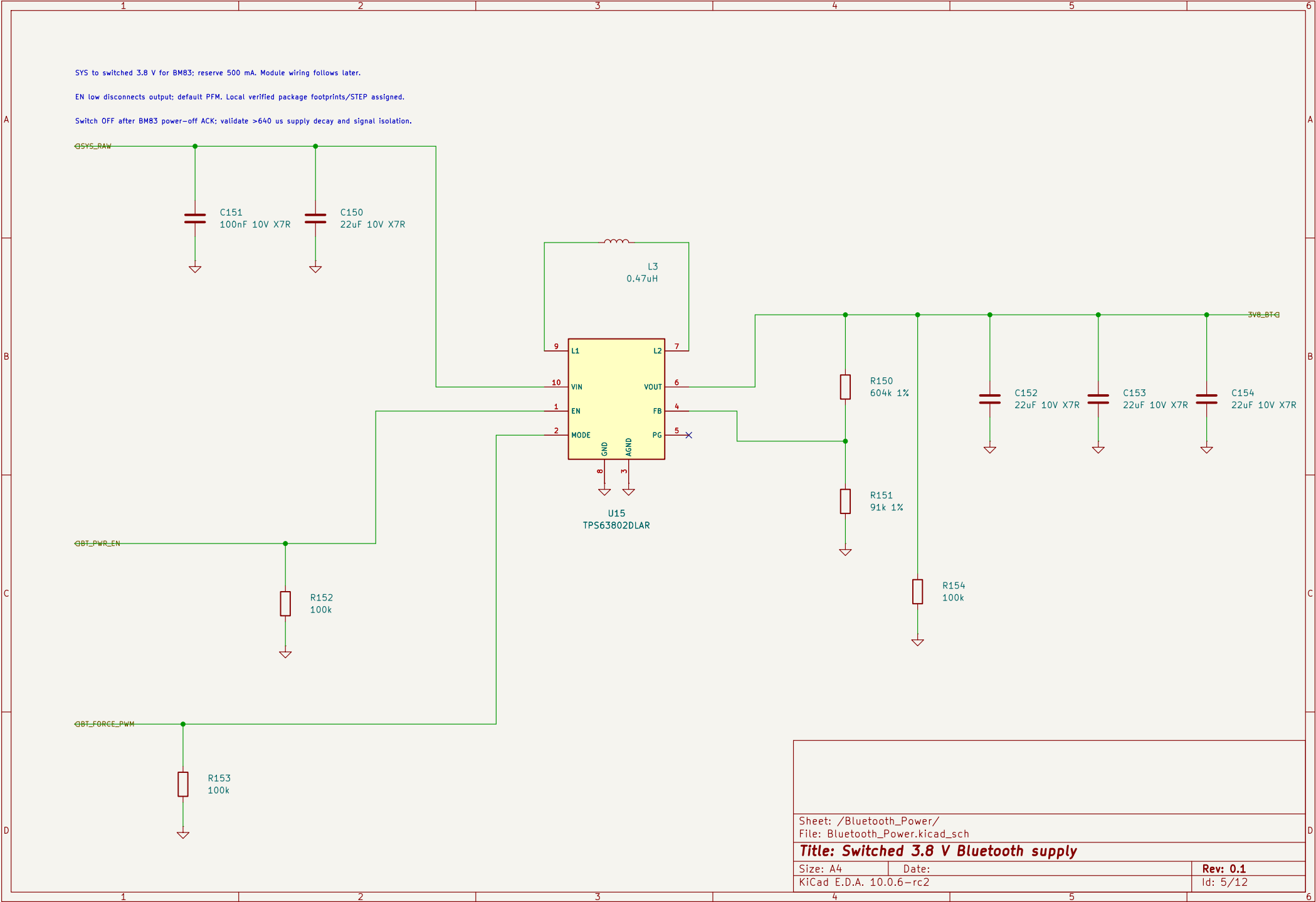
Size: A4

Date:

KiCad E.D.A. 10.0.6-rc2

Rev: 0.1

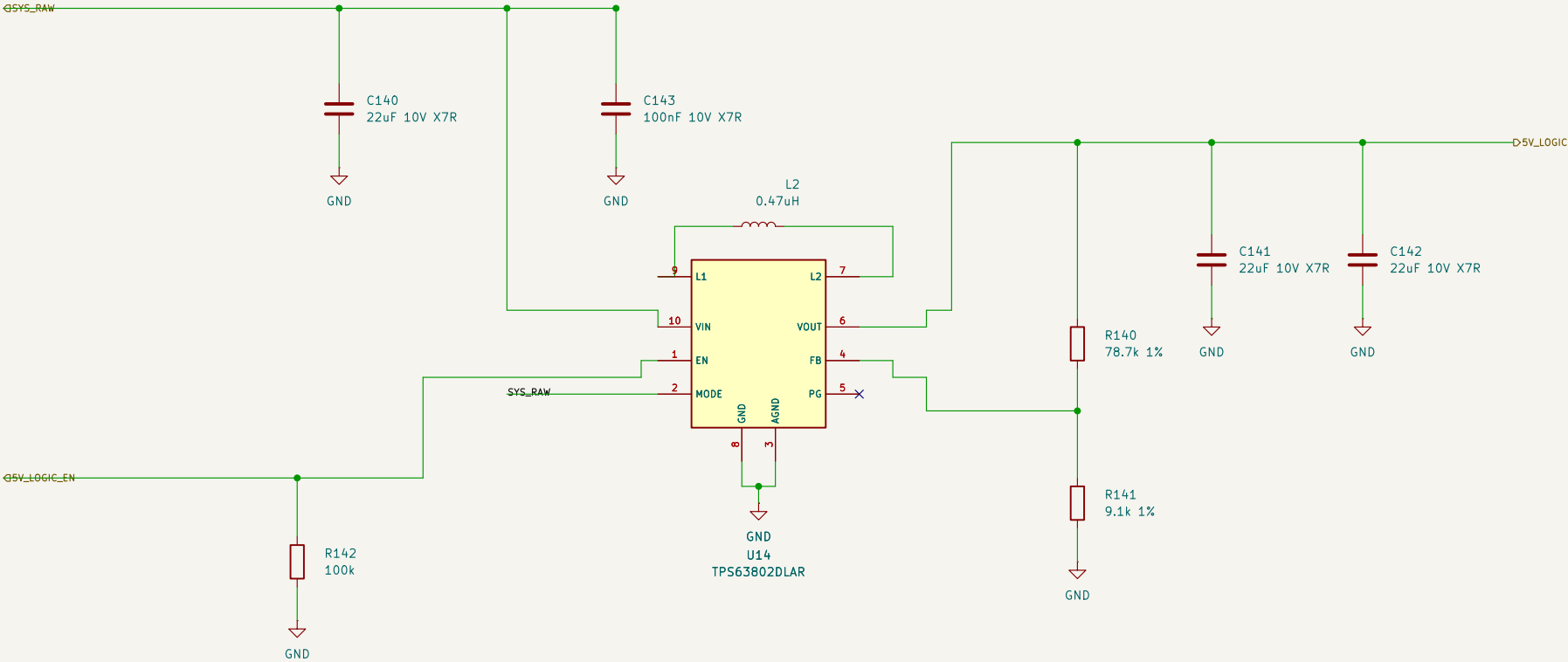
Id: 4/12



Protected 1S SYS to switched 5 V; reserve 500 mA total. Codec and other 5 V loads connect later.

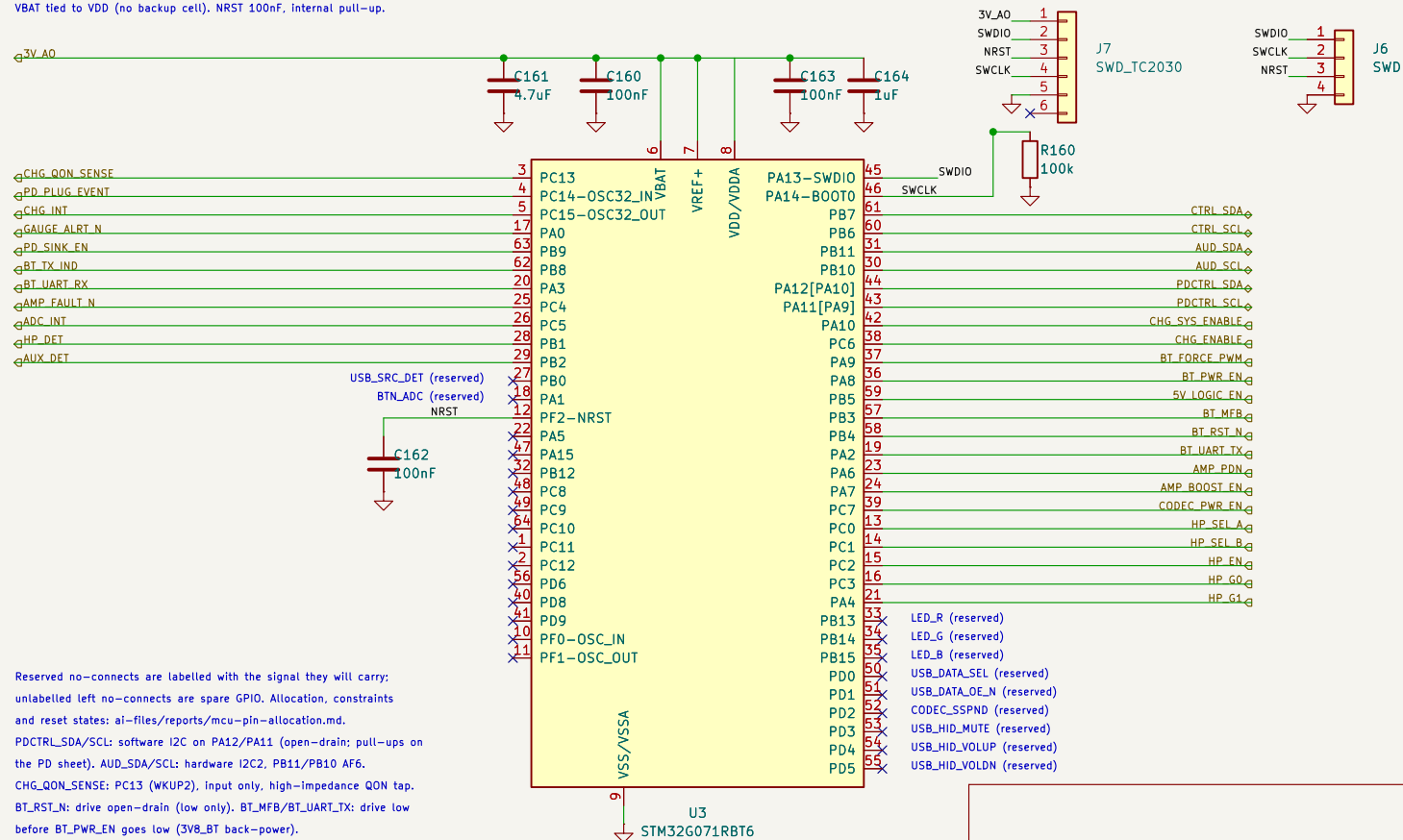
EN low = true input/output disconnect; MODE tied to SYS_RAW forces PWM after soft-start.

Steady PWM divider DC bound is within PCM2902C limits; soft-start uses PFM. Ripple and startup transient require qualification.



3V_AO supply: VDD/VDDA 100nF + 4.7uF (DS12232 fig. 13). VREF+ 100nF + 1uF (VREFBUF off, VREF+ = VDDA).
VBAT tied to VDD (no backup cell), NRST 100nF, internal pull-up.

SWD: J7 Tag-Connect TC2030-IDC (with legs: 6 SWO unused) and J6 4-pin 2.54 mm header share SWDIO, SWCLK, NRST
R160 holds PA14/BOOT0 low (nBOOT_SEL=1 ignores the pin)



Sheet: /MCU/
File: MCU.kicad_sch

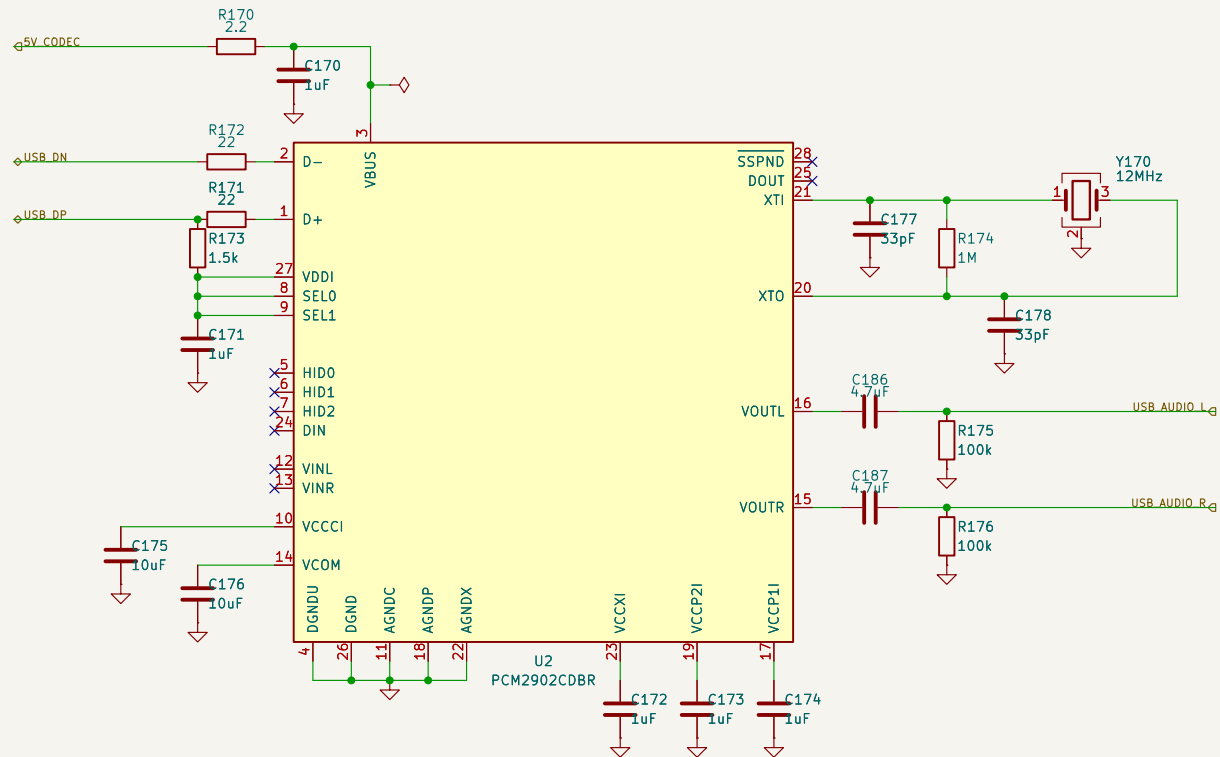
Title: MCU and programming

Size: A4
KiCad E.D.A. 10.0.6-rc2

Date:

Rev: 0.1

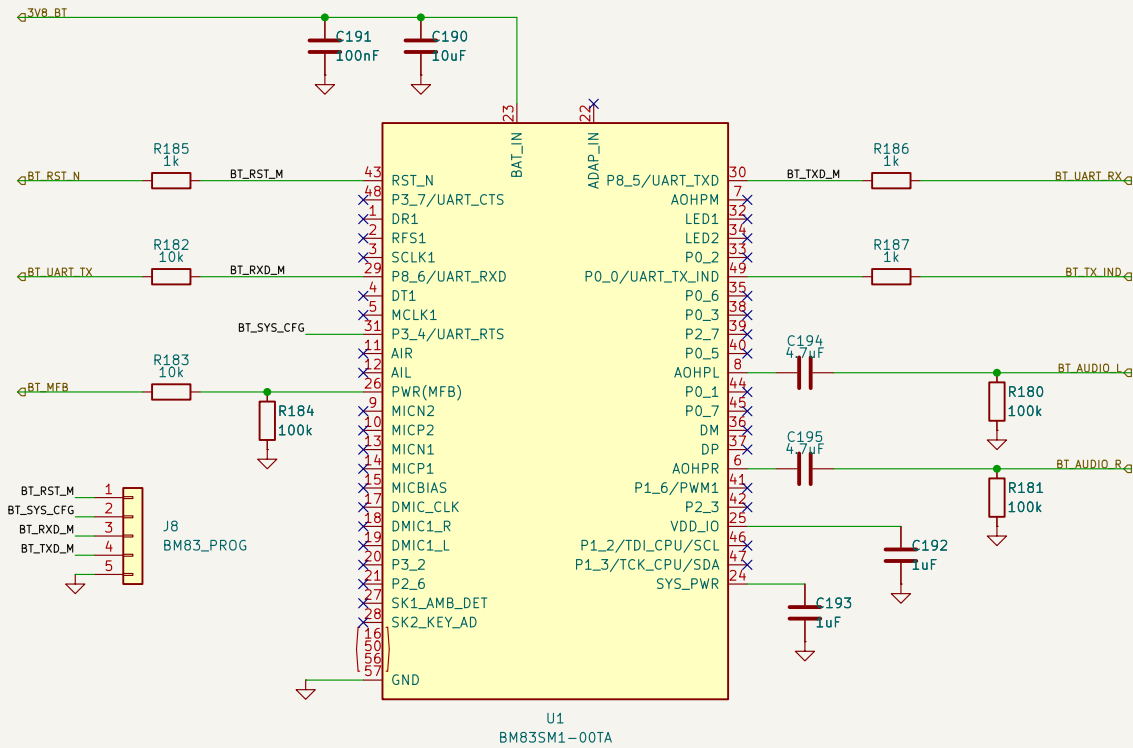
Id: 7/12



PCM2902C bus-powered codec from 5V_LOGIC; USB D+/D- path; analogue out to USB_AUD0_L/R

Title: *USB audio*

Rev: 0.1
Id: 8/12



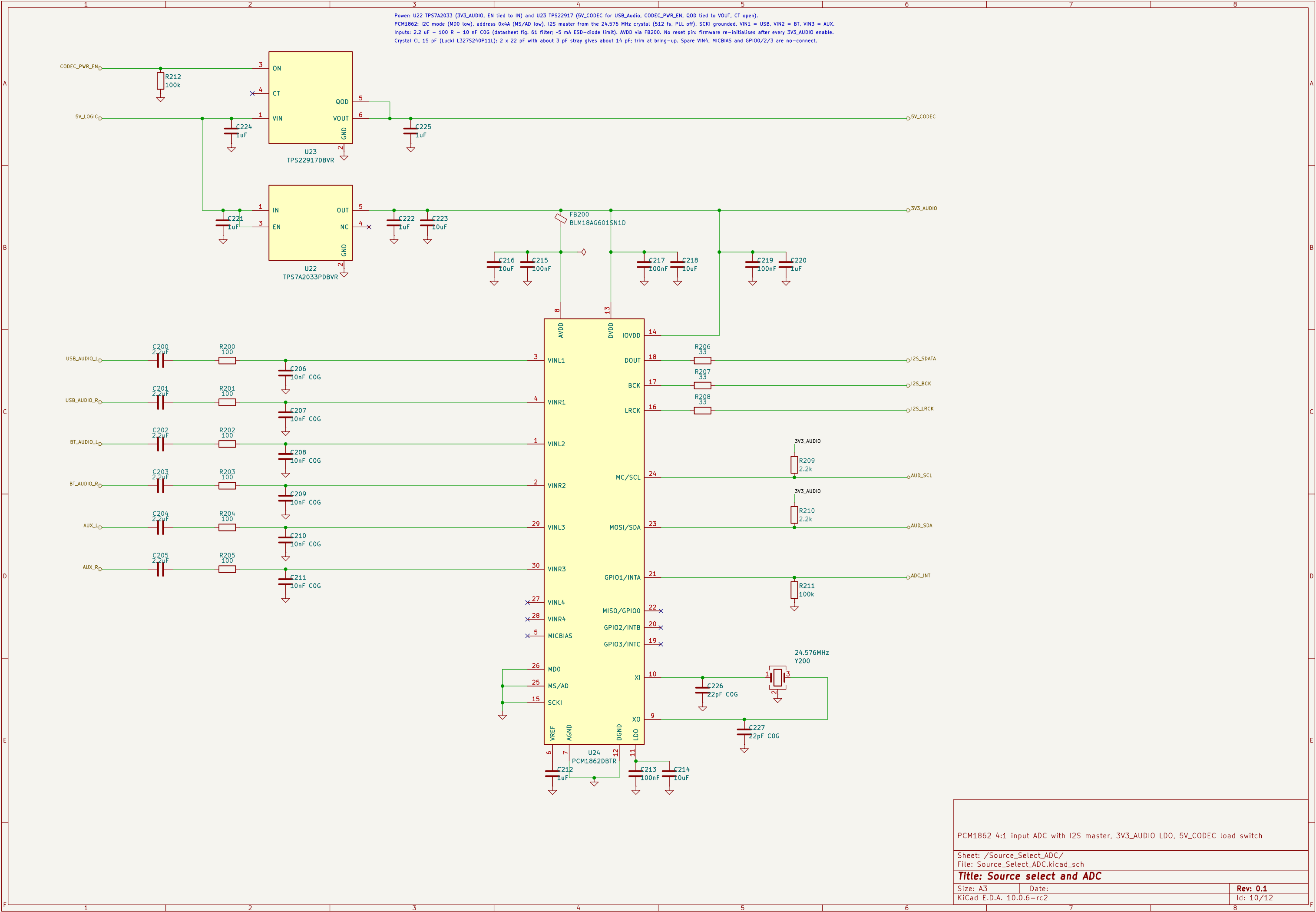
BM83SM1-00TA (DS70005402D). BAT_IN 3.2-4.2 V from 3V8_BT (U15); SYS_PWR/VDD_IO are outputs (decoupling only, do not load or feed). ADAP_IN unused (Internal charger/USB DFU not used).
Host mode UART from the MCU: PB_6 RXD / PB_5 TXD, P0_0 = UART_TX_IND (wakes the MCU). MFB (PWR) = power-on/wake key. RST_N = reset (active low, internal pull-up). Config Tool must set Host mode, P0_0 as TX_IND.
Back-power: MCU lines pass 10k/1k series resistors; firmware must drive PA2/PB3 low (PB4 open-drain/hi-Z) and disable BT_PWR_EN only after the module power-off ACK. Module outputs are 3.0-3.6 V into 5 V tolerant MCU pins.
Audio: single-ended DAC out (AOHPL/AOHPR) AC-coupled, 4.7 uF + 100k bleed, 0.74 Vrms full scale; AOHPM (capless sense) unused. Ports feed the later source-select sheet.
Antenna: integrated PCB antenna, no external RF parts. Keep the antenna end of the module off the board (no copper/components/keep-out) and metal 15 mm clear; see ai-files/reports/bluetooth-notes.md.
J8 programming/test (isUpdate/Config Tool): 1 RST_N, 2 P3_4 SYS_CFG (low at reset = test mode), 3 UART_RXD (adapter TX), 4 UART_TXD (adapter RX), 5 GND. 3V8_BT must be enabled first.

BM83 module: host UART, MFB/RST_N, analogue out

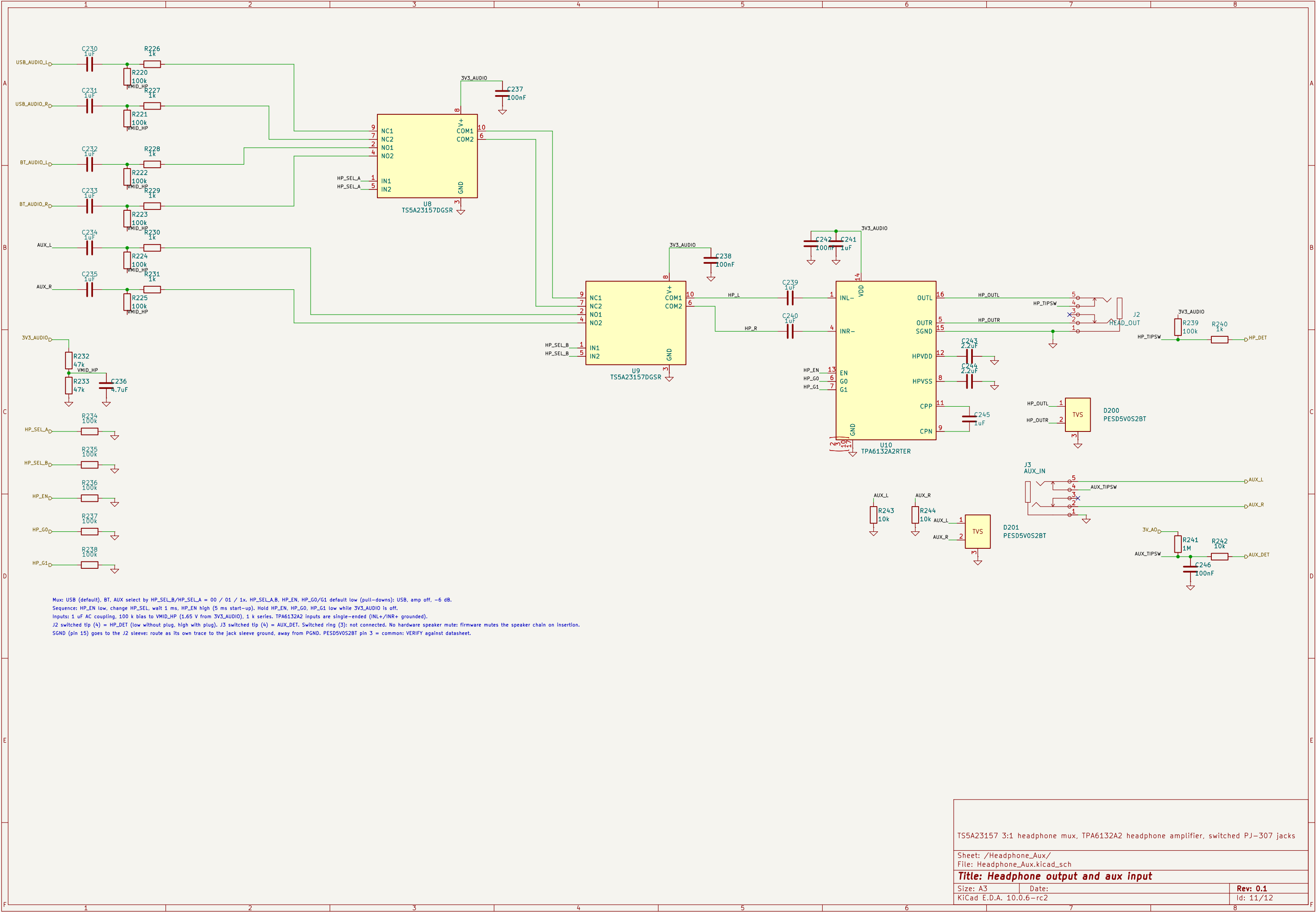
Sheet: /Bluetooth/
File: Bluetooth.kicad_sch

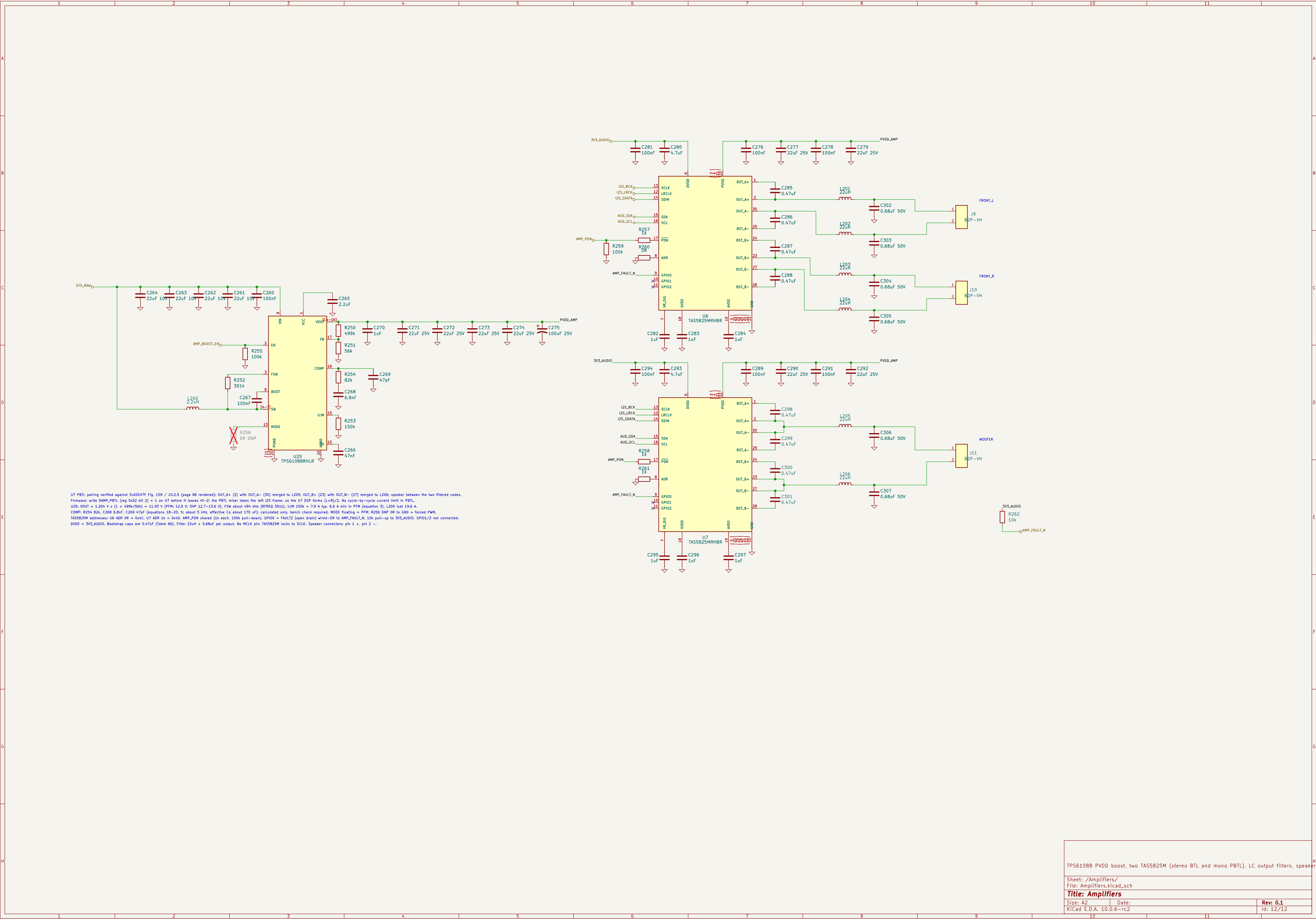
Title: Bluetooth audio

Size: A3	Date:	Rev: 0.1
KiCad E.D.A. 10.0.6-rc2		Id: 9/12



PCM1862 4:1 input ADC with I2S master, 3V3_AUDIO LDO, 5V_CODEC load switch





U7 PBTl pairing verified against SLASEH7F Fig. 159 / 10.2.5 (page 86 rendered): OUT_A+ (2) with OUT_A- (30) merged to L205; OUT_B+ (23) with OUT_B- (27) merged to L206; speaker between the two filtered nodes.
Firmware: write DAMP_PBTl (reg 0x02 bit 2) = 1 on U7 before it leaves HI-Z; the PBTl mixer takes the left I2S frame, so the U7 DSP forms (L+R)/2. No cycle-by-cycle current limit in PBTl.
U25: VOUT = 1.204 V × (1 + 499k/56k) = 11.93 V (PFM: 12.0 V; OVP 12.7-13.6 V); FSW about 49k Hz (RFREQ 301k); ILIM 150k = 7.9 A typ, 6.6 A min in PFM (equation 3); L200 Isat 19.6 A.
COMP: R254 82k, C268 6.8nF, C269 47pF (equations 18-20, fC about 5 kHz; effective Co about 170 uF); calculated only, bench check required. MODE floating = PFM; R256 DNP OR to GND = forced PWM.
TAS5825M addresses: U6 ADR OR = 0x4C, U7 ADR 1k = 0x4D, AMP_PDN shared (1k each, 100k pull-down), GPIO0 = FAULT2 (open drain) wired-OR to AMP_FAULT_N, 10k pull-up to 3V3_AUDIO, GPIO1/2 not connected.
DVDD = 3V3_AUDIO. Bootstrap caps are 0.47uF (Table 66). Filter 22uH + 0.68uF per output. No MCLK pin: TAS5825M locks to SCLK. Speaker connectors: pin 1 +, pin 2 -.